

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

KINETIC THEORY

LECTURE-03 **PHYSICS**

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

- 1** Revision ✓
- 2** Degrees of freedom ✓
- 3** Law of equi-partition of energy ✓
- 4** Application to specific heat capacities of gases

Recap from 1st Lecture

1 KTG

1
2
3
4
5
6
7
8
9
10



$$P = \frac{1}{3} \rho v^2$$

$$P = \frac{2}{3} \frac{K.E}{V}$$

Pressure and K.E
Relationship

$$K.E = \frac{3}{2} nRT$$

K.E and Temperature

Mean Free path

Recap from 2nd Lecture

Gaseous law

$$P \propto \frac{1}{V} \text{ (Boyle's law)}$$

$$P \propto T \text{ (Gay-Lussac law)}$$

$$V \propto T \text{ (Charles law)}$$

$$V \propto n \text{ (Avogadro's law)}$$

Ideal gas equation

$$PV = nRT$$

$$\begin{aligned} 1 \text{ mole} &= 6.023 \times 10^{23} \\ &\approx 6 \times 10^{23} \end{aligned}$$

Mean/Average velocity of gas molecules

$$V_a = \frac{V_1 + V_2 + V_3 + \dots + V_n}{n} = \sqrt{\frac{8RT}{\pi m}}$$

Root mean Square velocity of g m

$$V_{rms} = \sqrt{\frac{V_1^2 + V_2^2 + V_3^2 + \dots + V_n^2}{n}} = \sqrt{\frac{3RT}{m}}$$

Most Probable velocity of gas molecule

$$V_{mpv} = \sqrt{\frac{2RT}{m}}$$

Degree of freedom_n

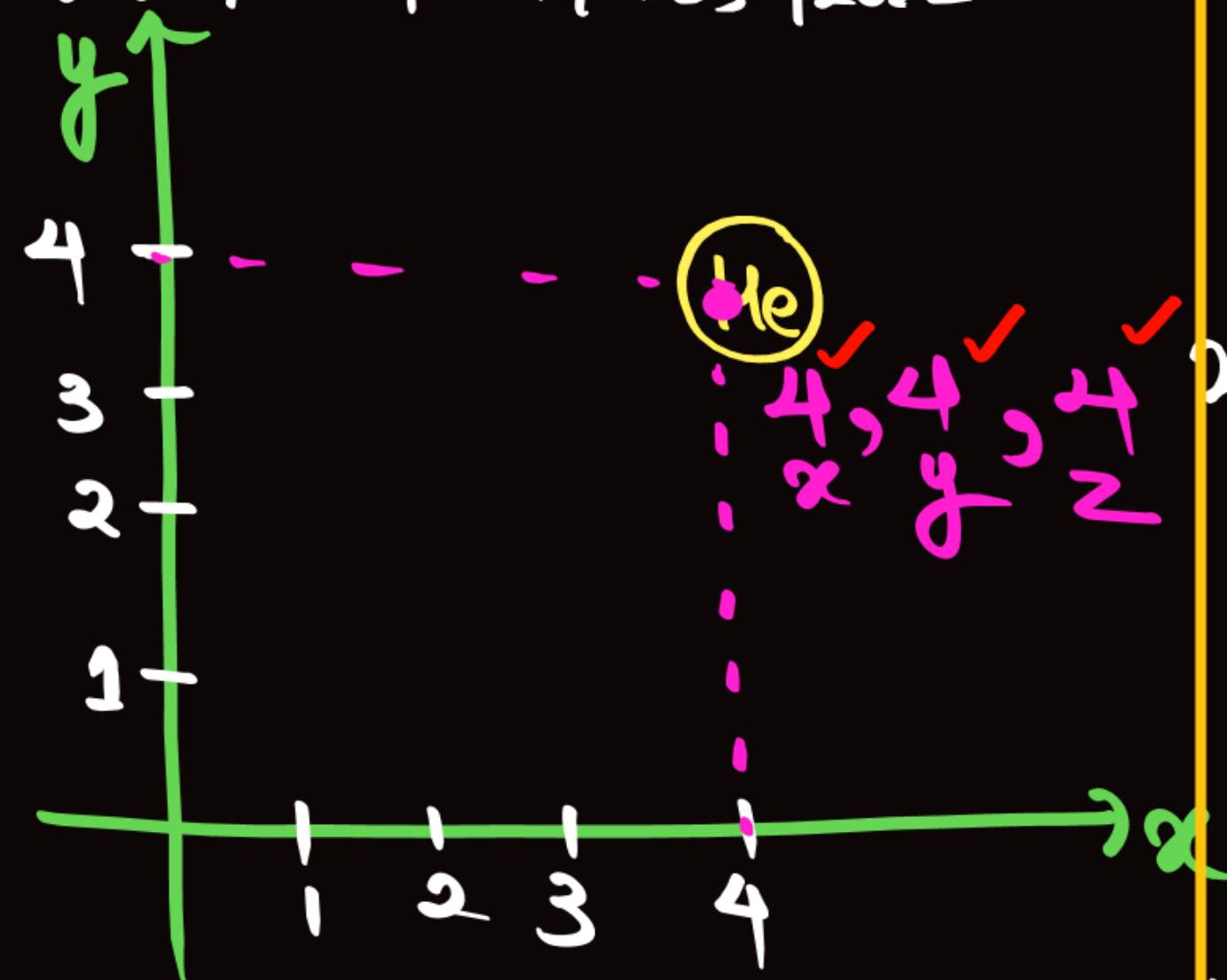
the number of co-ordinates required to specify the position of gas molecule

3 5 6

lecture no. 3

2 mark

2026 | 2025 | 2024 | 2023 | 2022



Type of Gas

Monatomic gas
(He, Ne, Ar)

Degree of Freedom
= 3 (1)

2024
✓ Diatomic gas
O₂ Cl₂
N₂

= 5

✓ Triatomic gas
O₃

= 6

Q The degree of freedom (n) for diatomic gas molecule ^②
(2024 JKBOSE)



(a) 1

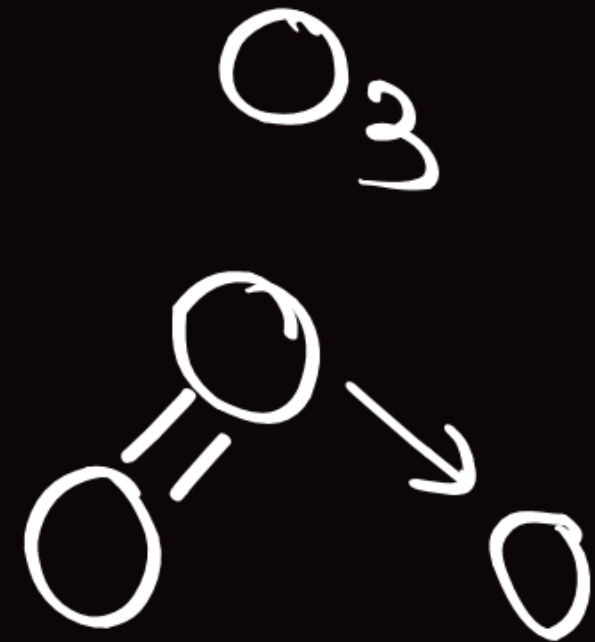
(Neet - 2026)

(b) 5 ✓

(Neet - 2023)

(c) 3

(d) 4



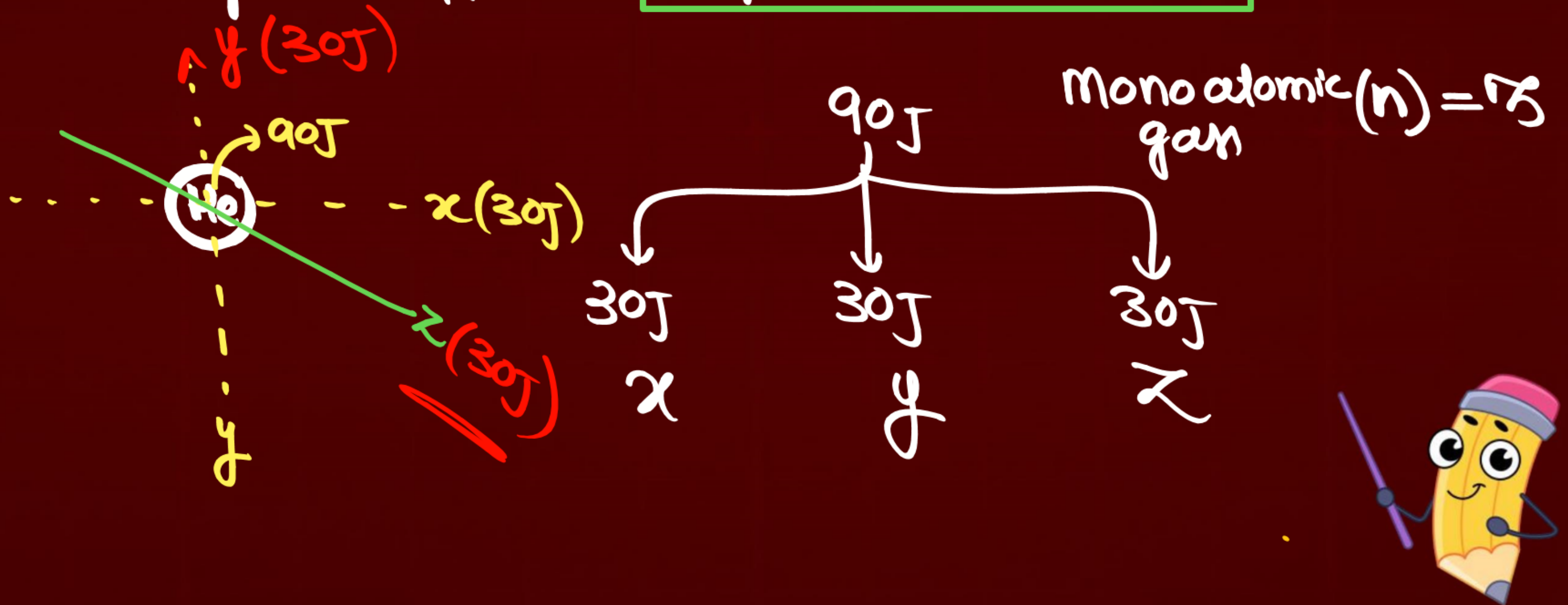
Law of equi-partition of Energy

(2 mark)

3



(•) The total Energy of gas molecule is equally distributed among the different degree of freedom n



Avagadro's Number

4



the number is given by Avagadro's

$$N_A = 6.023 \times 10^{23} = 1 \text{ mole}$$

board

Q 1 mole is equal to (2024)

~~(a) 6.023×10^{23}~~

(c) 3.023×10^{23}

(b) 7.842×10^{23}

(d) 5.67×10^{23}



Application of specific heat capacity

⑤

Specific heat is the amount of heat Required to Raised change the temp of 1gm substance by 1°C
unit ↗

$$Q = m S \Delta T$$

$$S \propto \frac{1}{\Delta T}$$

heat ↖

$$\frac{Q}{m \Delta T} = S$$

1gm 1°C ↓↓↓

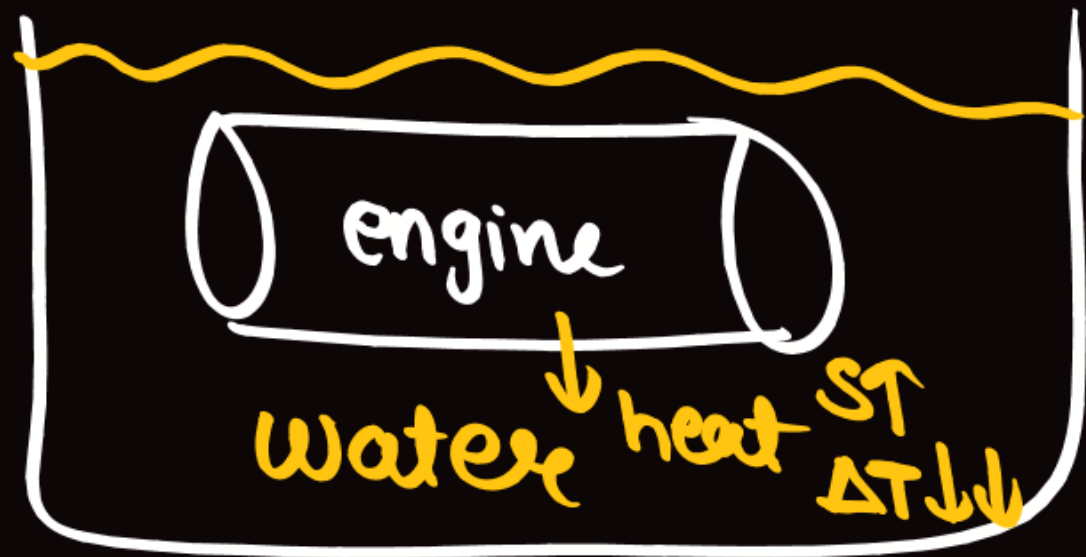
↑↑↑ Specific heat

(a) water is used as coolant

water have highest Specific heat

So ΔT low $\rightarrow$ late garam hota hai

$$\uparrow\uparrow S \propto \frac{1}{\Delta T \downarrow\downarrow}$$



(b) Metals have low S

Aluminium
Copper
 $\Rightarrow$ utensil are made from metal as metal have low Specific heat and show large change in Temp $\uparrow\uparrow$

$$\downarrow\downarrow S \propto \frac{1}{\Delta T \uparrow\uparrow}$$

$\downarrow$ Jaldi garam hote hai

⑥

Work done by compressing the Gas



Thermodynamics

$$W = -P \Delta V \text{ (Isobaric)}$$

$$W = -2.303 nRT \log \frac{V_2}{V_1} \text{ (Iso thermal)}$$

$$W = -\frac{R}{(\gamma-1)} (T_1 - T_2)$$



SUMMARY

- ✓ Degree of Freedom(n)
- ✓ law of $e-p$ - Energy
- ✓ N_A
- ✓ Application of specific heat

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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Thank

You

$$\frac{3 \times 60}{4}$$

